

PM SPE ENGINEERING CHANGE REQUEST (ECR) (Vendors – Complete Blocks 2 – 19 and Submit to PM SPE)		1. PM SPE CONTROL NUMBER ECP 23-11131130E5	
2. ORIGINATOR		3. ECR TYPE ☒ ECP ☐ RFD ☐ RFW	
a. TYPED NAME (First, Middle Initial, Last) Timothy M. Horne	b. ADDRESS (Street, City, State, Zip Code) 10125 Kingman Road, BLDG 317 Fort Belvoir, VA 22060	4. CLASSIFICATION ☐ Critical ☐ Minor ☒ Major	
5. TITLE OF ECR Alternative Use of Elongated FSP for MCEP Ballistic Testing			
6. ITEM DESIGNATION		7. PROCUREMENT ACTIVITY LMD, DLA, PdM SPE	8. BASELINE AFFECTED ☒ Functional ☐ Product ☐ Allocated
a. MCN / NSN All Eyewear - APEL/MCEP	b. MODEL / DESIGN CODE All Eyewear - APEL/MCEP		
9. END ITEM NOMENCLATURE All Eyewear - APEL/MCEP		10. CONTRACT NUMBER AND LINE ITEM MCEP APEL® requisitions	
11. LOTS / MATERIAL AFFECTED N/A		12. REOCCURRING DEVIATION (if Yes, Provide CAR# in Block 12 & Status in Block 18) N/A	
13. EFFECT ON COST / PRICE No		14. EFFECT ON DELIVERY SCHEDULE	
15. OTHER SYSTEMS / CONFIGURATION ITEMS / INTERFACES AFFECTED All Eyewear without any impacts			
16. DESCRIPTION OF ECR Update MIL-PRF-32432A to allow the use of both regular FSP and Elongated FSP for ballistic testing.			
17. NEED FOR ECR Vendors and test facilities have had troubles sourcing regular FSP and also controlling projectile tumble during impact. This ECP will allow vendors to use either regular FSP or elongating FSP based on their preference.			
18. CORRECTIVE ACTION TAKEN N/A			
19. SUBMITTING ACTIVITY			
a. TYPED NAME (FIRST, MIDDLE INITIAL, LAST) Timothy M. Horne		c. SIGNATURE HORNE.TIMOTHY.MAR TIN.1528909465	
b. TITLE / ORGANIZATION Product Engineer, PdM SPE		d. DATE (YYYYMMDD) 20231113	

Digitally signed by
HORNE.TIMOTHY.MARTIN.1528909465
Date: 2023.11.13 12:45:51 -05'00'

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20. PM SPE JUSTIFICATION In accordance with MIL-PRF-32432A, Paragraph 4.8.4 Ballistic fragmentation protection, Class 1 and Paragraph 4.8.5 Ballistic fragmentation protection, Class 2 and 3, projectiles shall be a fragment simulating projectile (FSP) of shape and dimensions as specified in Figure 3 and Figure 4 respectively. An excerpt of the requirement as well as the Engineering Drawings are attached. When tested, projectiles must meet the requirements set in 4.8.3 Projectile obliquity and yaw determination. "Projectile yaw shall be checked after every shot by performing a visual inspection of the impact location on the test item. A shot shall be "valid" if at least three (3) out of four (4) points of the projectile "chisel" are visible at the point of impact." Eyewear test laboratories expressed troubles in reliably of obtaining a valid shot using the standard FSP. Some vendors also expressed issues in sourcing standard FSP for testing. Due to one or both reasons some vendors independently made the decision to switch to an elongated FSP. An elongated FSP maintains weight requirements of the FSP and lengthens the FSP by sacrificing diameter of the projectile. This also creates a hollow tail on the back end of the FSP. ATC conducted comparison tests between the standard FSP and the elongated FSP. The findings are outlined in the Final Report for the Validation of the Elongated .15 and .22 Caliber Fragment-Simulating Projectiles (FSPs) ATEC Project No. 2021-DT-ATC-SOLSP-H9841 Report No. ATC-13243. These tests conducted V50 testing on various APEL spectacles and goggles in order to compare the projectile stability and the impact strength. The test report had 2 major conclusions: 1) The elongated FSP had less yaw than the standard FSP for both .15 caliber and .22 caliber. This means the impact was more direct and the elongated FSP are more stable in flight time. 2) The elongated FSP had a lower V50 velocity than the standard FSP for both .15 caliber and .22 caliber. A lower V50 translates that using an elongated FSP means it is a more difficult to pass a V0 RTP. The impact from an elongated projectile would be more likely to penetrate at a lower velocity than a standard FSP. This is likely because the elongated FSP is able to impact a lens closer to zero obliquity. PdM SPE has concluded that use of the elongated projectiles is a more strenuous test while decreasing misfires due to projectile yaw. Should test labs wish to utilize the elongated FSP, they may be subjected to availability.		
21. ATTACHMENTS 1) FSP Requirements and Drawing Except from MIL-PRF-32432A 2) Report No. ATC-13242		
22. COR (APPROVAL AUTHORITY FOR MINOR ECRS) / PM SPE TECHNICAL STAFF APPROVAL / DISAPPROVAL FOR MAJOR OR CRITICAL ECRS)		
a. APPROVAL ☒ Approved ☐ Disapproved	b. SIGNATURE FLOYD.WHITNEY.HIER HOLZER.1282204602 Digitally signed by FLOYD.WHITNEY.HIERHOLZER.1282204602 Date: 2023.11.13 18:09:38 -05'00'	c. DATE SIGNED (YYYYMMDD) 20231113
23. APM APPROVAL / DISAPPROVAL (APPROVAL AUTHORITY FOR MAJOR ECRs)		
a. APPROVAL ☒ Approved ☐ Disapproved	b. SIGNATURE NULK.MATTHEW.PAT RICK.1020418970 Digitally signed by NULK.MATTHEW.PATRICK.1020418970 Date: 2023.11.13 20:16:58 -05'00'	c. DATE SIGNED (YYYYMMDD) 20231113
24. PM SPE APPROVAL / DISAPPROVAL (APPROVAL AUTHORITY FOR CRITICAL ECRs)		
a. APPROVAL ☐ Approved ☐ Disapproved	b. SIGNATURE	c. DATE SIGNED (YYYYMMDD)

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